Course Code: 2165119 **Course Name:** ASD&D Lab

Class: TE **Batch:** T23 **Semester:** V

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EXPERIMENT NO. 05

Aim: - To automate application deployment using GitHub and Jenkins.

Theory:

Continuous Integration and Continuous Deployment (CI/CD) pipelines automate the process of software delivery. **Jenkins** is a widely used open-source automation server that facilitates this by building, testing, and deploying code every time a change is made.

This experiment utilizes a **Declarative Jenkins Pipeline**, which defines pipeline, agent, stages, and steps as its core syntax elements. The standard automated pipeline consists of three main stages:

1. **Build:** This stage prepares and creates the application. For a compiled language, it compiles source code; for this experiment's HTML application, no complex compilation is required.
2. **Test:** This stage checks whether the application works correctly, running automated test scripts if configured.
3. **Deploy:** This final stage puts the successfully tested application into its target environment.

Webhooks play a crucial role in full automation. A webhook is an HTTP callback that occurs when something happens. The ultimate goal of this automated architecture is a seamless flow: a developer changes code, pushes it to GitHub, GitHub triggers a Webhook, the Webhook sends a payload to Jenkins, and Jenkins automatically executes the Build, Test, and Deploy stages to update the application without manual intervention.

IMPLEMENTATION

Step 1: Setting up the GitHub Repository and Application Code

A GitHub repository (ASDD_Exp5_Webhook) was created and cloned to the local environment. A sample index.html application file and a Jenkinsfile were created. The Jenkinsfile contained the declarative script defining the Build, Test, and Deploy stages.

```
PS D:\ASDD> git clone https://github.com/SwarajKanse/ASDD_Exp5_Webhook.git
PS D:\ASDD> cd ASDD_Exp5_Webhook
PS D:\ASDD\ASDD_Exp5_Webhook> Set-Content index.html "<h1>My Auto-Deployed App</h1>"
PS D:\ASDD\ASDD_Exp5_Webhook> notepad Jenkinsfile
PS D:\ASDD\ASDD_Exp5_Webhook> git add .
PS D:\ASDD\ASDD_Exp5_Webhook> git commit -m "Initial commit for Exp 5"
PS D:\ASDD\ASDD_Exp5_Webhook> git push origin main
```

Step 2: Configuring the Jenkins Pipeline

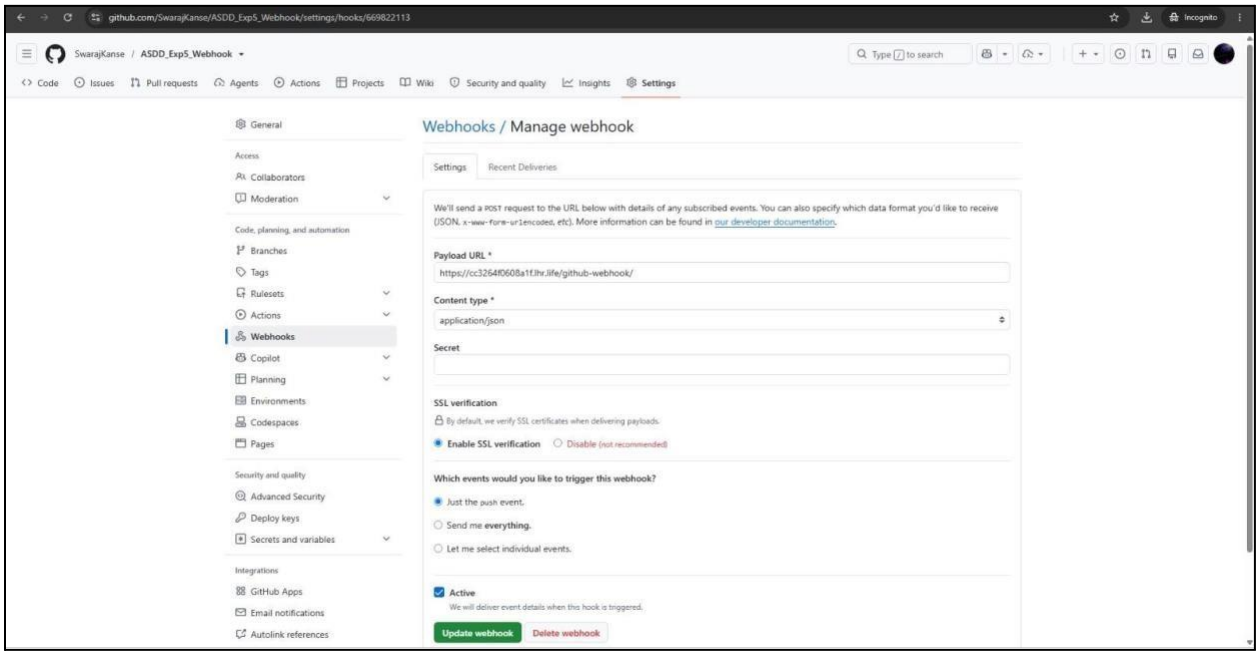
In the local Jenkins server (running on localhost:8081), a new Pipeline project named GitHubJenkins-Demo was created. Under the Build Triggers configuration, **"GitHub hook trigger for GITScm polling"** was enabled. The Pipeline definition was set to "Pipeline script from SCM", linked to the GitHub repository URL, and configured to look for the Jenkinsfile on the main branch.

Step 3: Creating a Public Tunnel and Configuring the GitHub Webhook

Because the local Jenkins server was running on a private lab network behind a firewall, GitHub's public servers could not reach it directly (resulting in a failed to connect to host error). To bypass this, an SSH tunnel (localhost.run) was used via PowerShell to generate a temporary public URL (https://cc3264f0608a1f.lhr.life) that forwarded traffic to localhost:8081.

```
PS C:\Users\Lab203> ssh -R 80:localhost:8081 nokey@localhost.run
```

This public URL was then added to the GitHub repository under **Settings > Webhooks** with the /github-webhook/ endpoint and application/json content type. The connection was established successfully.

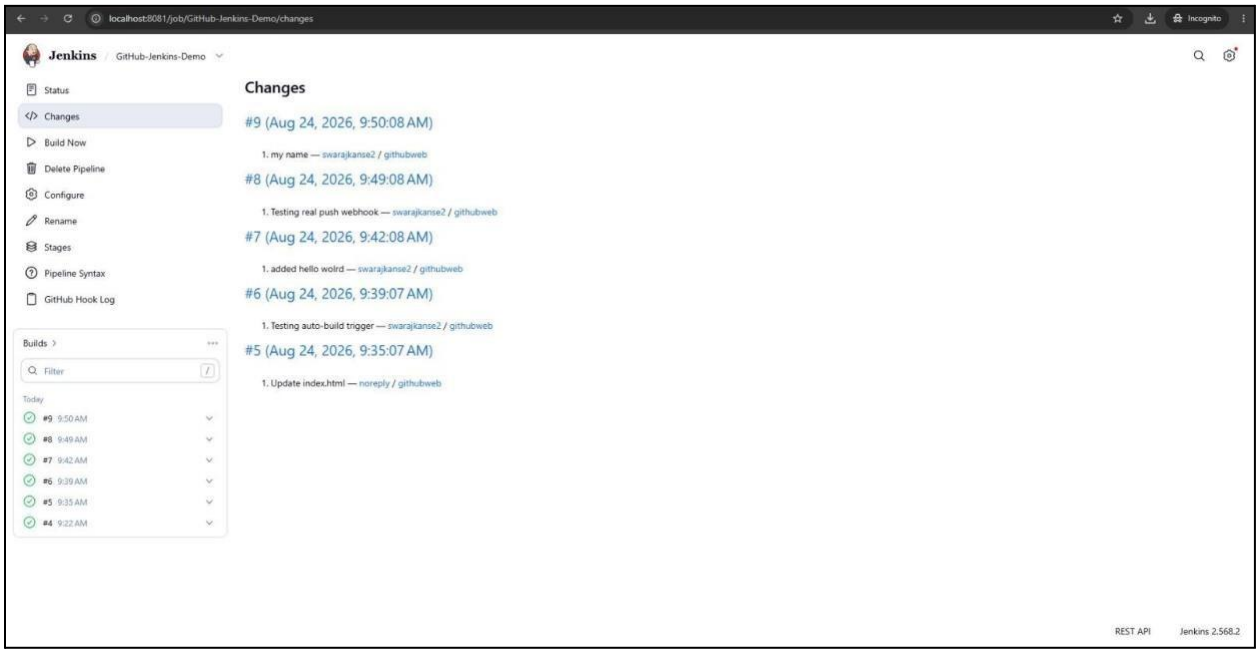


Step 4: Triggering and Verifying the Automated Build

To test the true automation (Push-to-Deploy), a modification was made to the local code and pushed to the remote repository.

```
PS D:\ASDD\ASDD_Exp5_Webhook> Set-Content index.html "<h1>True Webhook Push is Successful!</
PS D:\ASDD\ASDD_Exp5_Webhook> git add .
PS D:\ASDD\ASDD_Exp5_Webhook> git commit -m "Testing real push webhook"
PS D:\ASDD\ASDD_Exp5_Webhook> git push origin main
```

Immediately upon the push completing, GitHub fired the Webhook payload through the SSH tunnel to the Jenkins server. Observing the Jenkins dashboard confirmed that Build #8 ("Testing real push webhook") was triggered automatically without manual intervention, successfully executing the Build, Test, and Deploy stages.



Conclusion: Thus, we have successfully studied CI/CD pipelines and utilized Jenkins, GitHub, and Webhooks (with SSH tunneling) to create a fully automated application deployment workflow that triggers instantly upon code commits.